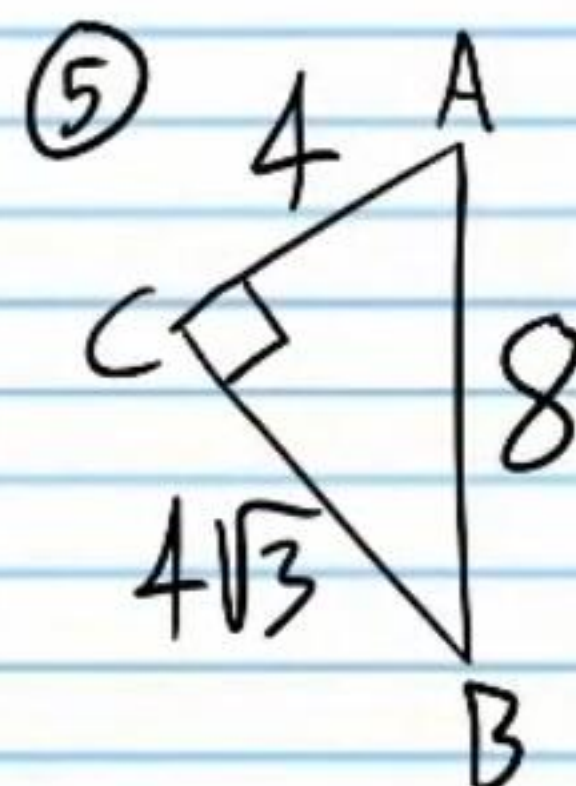
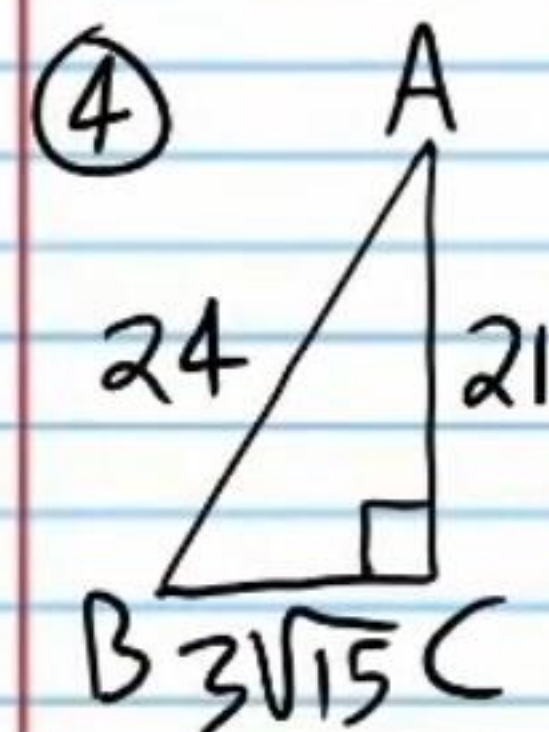
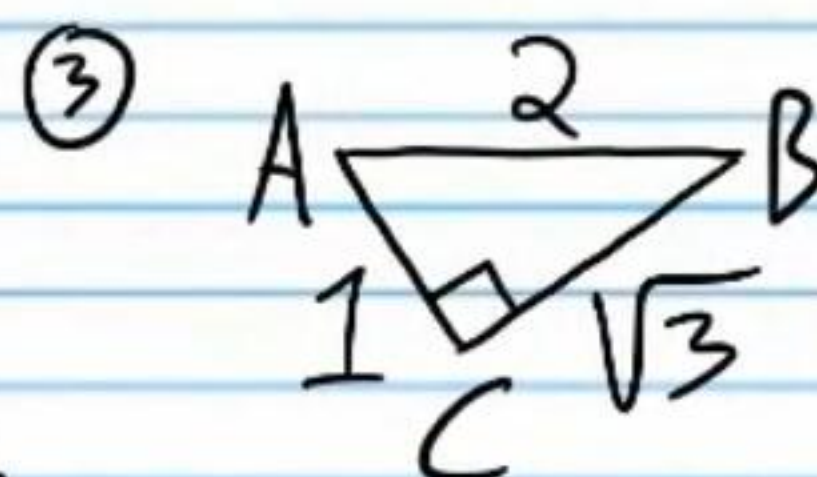
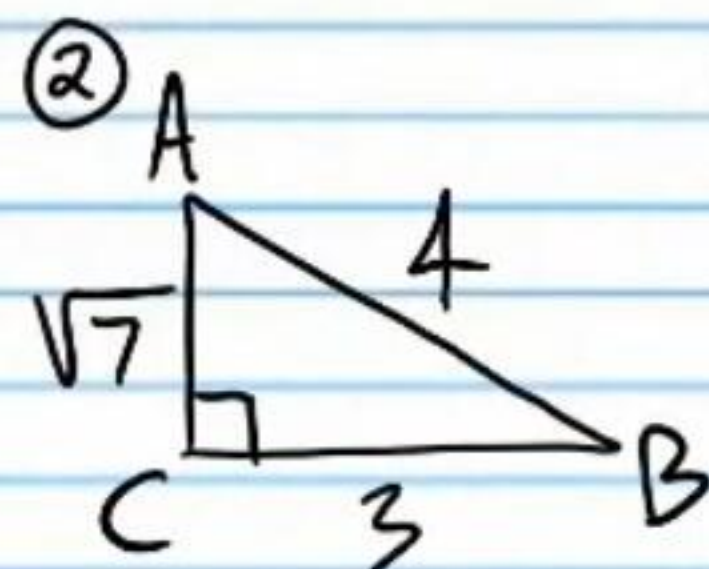
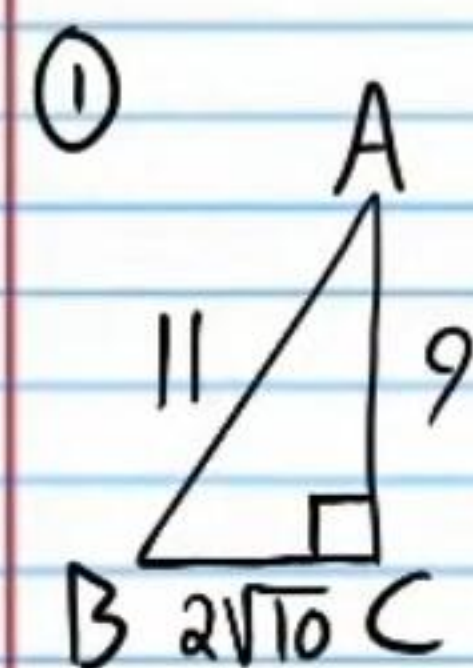
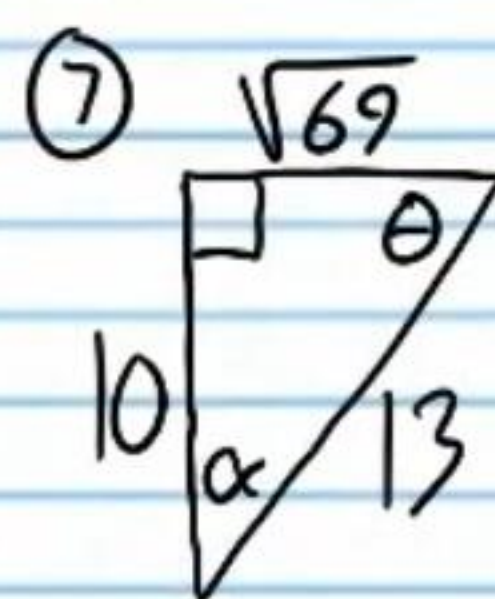
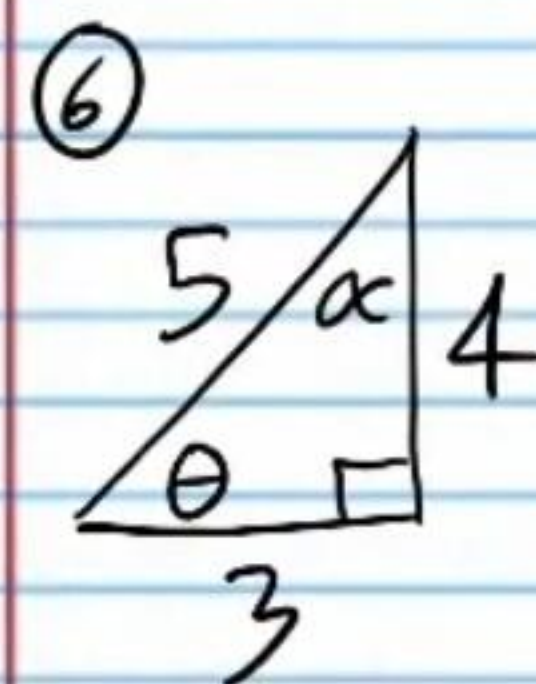


Right Triangles (Part II): The Sine, Cosine, and Tangent Ratios (Homework)

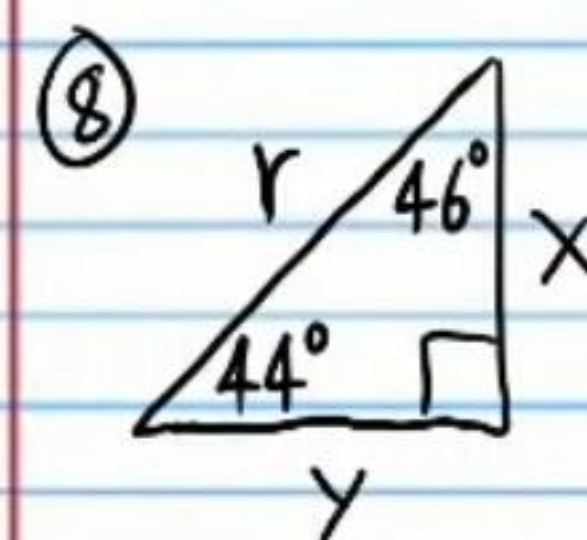
For the triangles below, find $\sin A$, $\cos A$, and $\tan A$. Then find $\sin B$, $\cos B$, and $\tan B$.



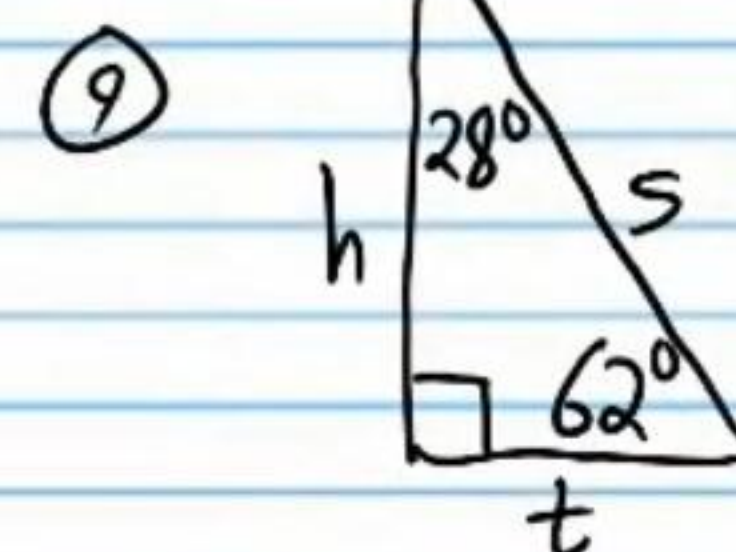
Find $\sin \theta$, $\cos \theta$, and $\tan \theta$. Then find $\sin \alpha$, $\cos \alpha$, and $\tan \alpha$.



Find the ratios listed below.



$\sin 46^\circ$, $\cos 46^\circ$, $\tan 46^\circ$,
 $\sin 44^\circ$, $\cos 44^\circ$, and $\tan 44^\circ$



$\sin 28^\circ$, $\cos 28^\circ$, $\tan 28^\circ$,
 $\sin 62^\circ$, $\cos 62^\circ$, and $\tan 62^\circ$

Use a calculator to find the sine, cosine, and tangent of each of the following angles. Round to four decimal places.

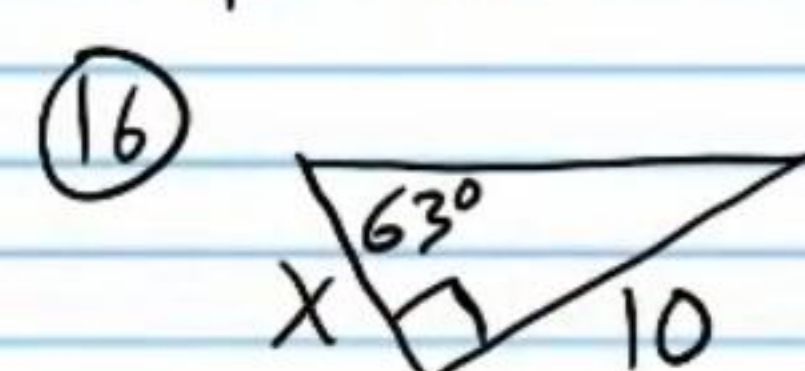
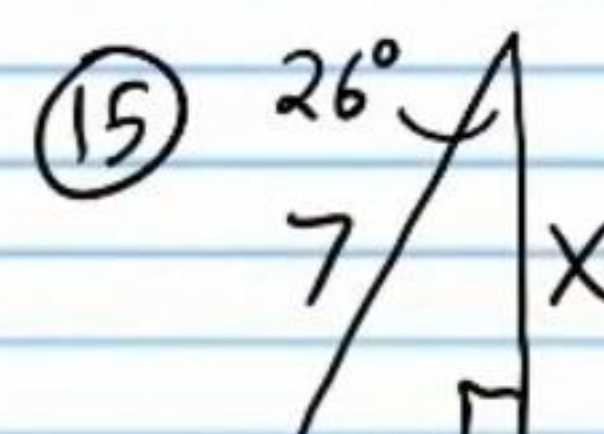
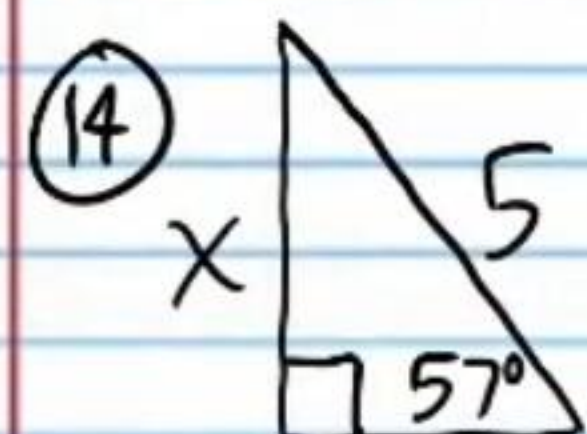
⑩ $\theta = 68^\circ$ ⑪ $\omega = 15^\circ$

Find the ratios below without a calculator.
Note: You may need to refer to the diagrams of special triangles shown in class.

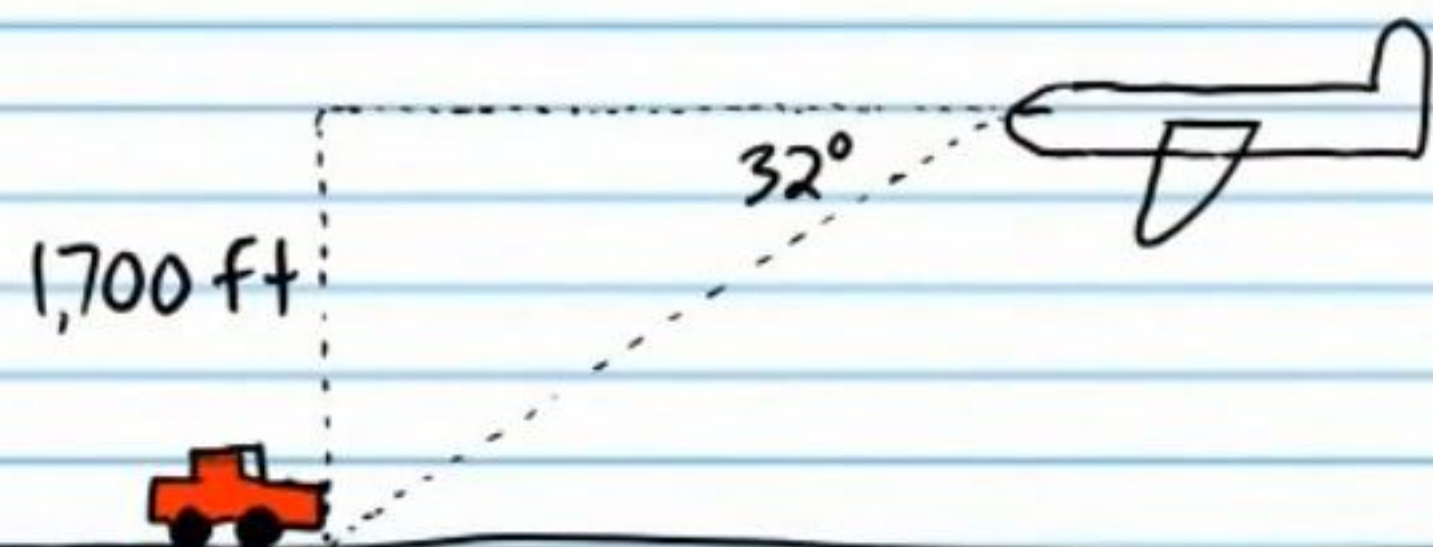
⑫ $\sin 30^\circ$
 $\cos 60^\circ$
 $\tan 45^\circ$

⑬ $\sin 60^\circ$
 $\cos 45^\circ$
 $\tan 30^\circ$

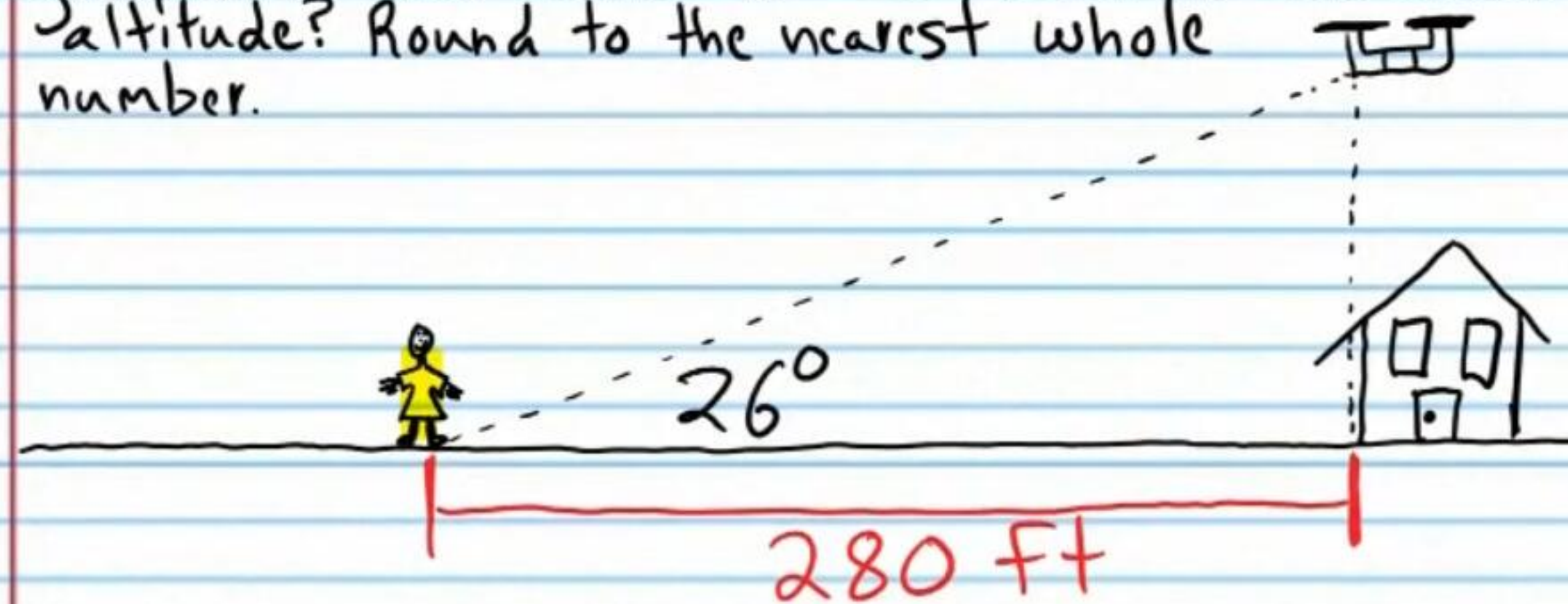
Find X. Round to four decimal places.



(17) If the angle of depression from an airplane pilot to a car is 32° , and the pilot is 1,700 feet in the air, what is the horizontal distance from the plane to the car? Round to the nearest whole number.



(18) Linda is flying her toy drone. She knows that the drone is directly above her house, which is 280 feet from where she is standing, and the angle of elevation from the ground to the drone is 26° . What is the drone's altitude? Round to the nearest whole number.



(19) Considering the previous problem, if Linda wants to fly the drone to where she is standing, what is the distance of the shortest path that the drone could take? Round to the nearest whole number.

Answers

$$\begin{aligned} \textcircled{1} \quad \sin A &= \frac{2\sqrt{10}}{11} & \sin B &= \frac{9}{11} \\ \cos A &= \frac{9}{11} & \cos B &= \frac{2\sqrt{10}}{11} \\ \tan A &= \frac{2\sqrt{10}}{9} & \tan B &= \frac{9\sqrt{10}}{20} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \sin A &= \frac{3}{4} & \sin B &= \frac{\sqrt{7}}{4} \\ \cos A &= \frac{\sqrt{7}}{4} & \cos B &= \frac{3}{4} \\ \tan A &= \frac{3\sqrt{7}}{7} & \tan B &= \frac{\sqrt{7}}{3} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \sin A &= \frac{\sqrt{3}}{2} & \sin B &= \frac{1}{2} \\ \cos A &= \frac{1}{2} & \cos B &= \frac{\sqrt{3}}{2} \\ \tan A &= \sqrt{3} & \tan B &= \frac{\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad & \star \text{ Reduce!} \\ \sin A &= \frac{\sqrt{15}}{8} & \sin B &= \frac{7}{8} \\ \cos A &= \frac{7}{8} & \cos B &= \frac{\sqrt{15}}{8} \\ \tan A &= \frac{\sqrt{15}}{7} & \tan B &= \frac{7\sqrt{15}}{15} \end{aligned}$$

$$\begin{aligned} \textcircled{5} \quad & \star \text{ Reduce!} \\ \sin A &= \frac{\sqrt{3}}{2} & \sin B &= \frac{1}{2} \\ \cos A &= \frac{1}{2} & \cos B &= \frac{\sqrt{3}}{2} \\ \tan A &= \sqrt{3} & \tan B &= \frac{\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad \sin \theta &= \frac{4}{5} & \sin \alpha &= \frac{3}{5} \\ \cos \theta &= \frac{3}{5} & \cos \alpha &= \frac{4}{5} \\ \tan \theta &= \frac{4}{3} & \tan \alpha &= \frac{3}{4} \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad \sin \theta &= \frac{10}{13} & \sin \alpha &= \frac{\sqrt{69}}{13} \\ \cos \theta &= \frac{\sqrt{69}}{13} & \cos \alpha &= \frac{10}{13} \\ \tan \theta &= \frac{10\sqrt{69}}{69} & \tan \alpha &= \frac{\sqrt{69}}{10} \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad \sin 46^\circ &= \frac{y}{r} & \sin 44^\circ &= \frac{x}{r} \\ \cos 46^\circ &= \frac{x}{r} & \cos 44^\circ &= \frac{y}{r} \\ \tan 46^\circ &= \frac{y}{x} & \tan 44^\circ &= \frac{x}{y} \end{aligned}$$

$$\begin{aligned} \textcircled{9} \quad \sin 28^\circ &= \frac{t}{5} & \sin 62^\circ &= \frac{h}{5} \\ \cos 28^\circ &= \frac{h}{5} & \cos 62^\circ &= \frac{t}{5} \\ \tan 28^\circ &= \frac{t}{h} & \tan 62^\circ &= \frac{h}{t} \end{aligned}$$

$$\begin{aligned} \textcircled{10} \quad \sin 68^\circ &\approx 0.9272 \\ \cos 68^\circ &\approx 0.3746 \\ \tan 68^\circ &\approx 2.4751 \end{aligned}$$

$$\begin{aligned} \textcircled{11} \quad \sin 15^\circ &\approx 0.2588 \\ \cos 15^\circ &\approx 0.9659 \\ \tan 15^\circ &\approx 0.2679 \end{aligned}$$

$$\begin{aligned} \textcircled{12} \quad \sin 30^\circ &= \frac{1}{2} \\ \cos 60^\circ &= \frac{1}{2} \\ \tan 45^\circ &= 1 \end{aligned}$$

$$\begin{aligned} \textcircled{13} \quad \sin 60^\circ &= \frac{\sqrt{3}}{2} \\ \cos 45^\circ &= \frac{\sqrt{2}}{2} \\ \tan 30^\circ &= \frac{\sqrt{3}}{3} \end{aligned}$$

$$\textcircled{14} \quad x \approx 4.1934 \quad \textcircled{15} \quad x \approx 6.2916 \quad \textcircled{16} \quad x \approx 5.0953$$

$$\textcircled{17} \quad 2,721 \text{ ft} \quad \textcircled{18} \quad 137 \text{ ft} \quad \textcircled{19} \quad 312 \text{ ft}$$